

Properties of Semiconductor Alloys: Group-IV, III–V and II–VI Semiconductors

SADAO ADACHI

Gunma University, Gunma, Japan



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Contents

Series Preface	xvii
Preface	xix
Abbreviations and Acronyms	xxi
Introductory Remarks	1
A.1 An Alloy and a Compound	1
A.2 Grimm–Sommerfeld Rule	2
A.3 An Interpolation Scheme	4
References	7
1 Structural Properties	9
1.1 Ionicity	9
1.2 Elemental Isotopic Abundance and Molecular Weight	9
1.3 Crystal Structure	11
1.3.1 Random Alloy	11
1.3.2 Spontaneous Ordering	11
(a) Group-IV Semiconductor Alloy	11
(b) III–V Semiconductor Alloy	14
(c) II–VI Semiconductor Alloy	15
1.4 Lattice Constant and Related Parameters	15
1.4.1 CuAu Alloy: Ordered and Disordered States	15
1.4.2 Non-alloyed Semiconductor	16
1.4.3 Semiconductor Alloy	19
(a) Group-IV Semiconductor	19
(b) III–V Semiconductor	22
(c) II–VI Semiconductor	29
1.5 Coherent Epitaxy and Strain Problem	32
1.5.1 Bilayer Model	32
1.5.2 Elastic Strain and Lattice Deformation	33
1.5.3 Critical Thickness	37
1.6 Structural Phase Transition	39
1.7 Cleavage Plane	41
1.7.1 Cleavage	41
1.7.2 Surface Energy	41
References	42

2 Thermal Properties	45
2.1 Melting Point and Related Parameters	45
2.1.1 Phase Diagram	45
(a) Group-IV Semiconductor Alloy	45
(b) III–V Semiconductor Alloy	45
(c) II–VI Semiconductor Alloy	48
2.1.2 Melting Point	51
2.2 Specific Heat	51
2.2.1 Group-IV Semiconductor Alloy	51
2.2.2 III–V Semiconductor Alloy	54
2.2.3 II–VI Semiconductor Alloy	56
2.3 Debye Temperature	56
2.3.1 General Considerations	56
2.3.2 Group-IV Semiconductor Alloy	57
2.3.3 III–V Semiconductor Alloy	58
2.3.4 II–VI Semiconductor Alloy	58
2.4 Thermal Expansion Coefficient	59
2.4.1 Group-IV Semiconductor Alloy	59
2.4.2 III–V Semiconductor Alloy	61
2.4.3 II–VI Semiconductor Alloy	63
2.5 Thermal Conductivity and Diffusivity	63
2.5.1 Thermal Conductivity	63
(a) General Considerations	63
(b) Group-IV Semiconductor Alloy	66
(c) III–V Semiconductor Alloy	68
(d) II–VI Semiconductor Alloy	74
2.5.2 Thermal Diffusivity	75
(a) General Considerations	75
(b) Alloy Value	75
References	76
 3 Elastic Properties	 81
3.1 Elastic Constant	81
3.1.1 General Considerations	81
3.1.2 Room-temperature Value	81
(a) Group-IV Semiconductor Alloy	81
(b) III–V Semiconductor Alloy	81
(c) II–VI Semiconductor Alloy	84
3.1.3 External Perturbation Effect	86
(a) Temperature Effect	86
(b) Pressure Effect	88
3.2 Third-order Elastic Constant	89
3.3 Young's Modulus, Poisson's Ratio and Similar Properties	89
3.3.1 Group-IV Semiconductor Alloy	89
3.3.2 III–V Semiconductor Alloy	89
3.3.3 II–VI Semiconductor Alloy	90

3.4	Microhardness	92
3.4.1	Group-IV Semiconductor Alloy	92
3.4.2	III–V Semiconductor Alloy	92
3.4.3	II–VI Semiconductor Alloy	93
3.5	Sound Velocity	96
	References	97
4	Lattice Dynamic Properties	99
4.1	Phonon Dispersion Relationships	99
4.2	Phonon Frequency	99
4.2.1	General Considerations	99
4.2.2	Room-temperature Value	100
	(a) Group-IV Semiconductor Alloy	100
	(b) III–V Semiconductor Alloy	104
	(c) II–VI Semiconductor Alloy	109
4.2.3	External Perturbation Effect	112
	(a) Group-IV Semiconductor Alloy	112
	(b) III–V Semiconductor Alloy	113
	(c) II–VI Semiconductor Alloy	115
4.3	Mode Grüneisen Parameter	119
4.3.1	Phonon Deformation Potential	121
	References	123
5	Collective Effects and Some Response Characteristics	125
5.1	Piezoelectric Constant	125
5.1.1	General Considerations	125
5.1.2	Alloy Value	125
5.2	Fröhlich Coupling Constant	129
5.2.1	General Considerations	129
5.2.2	Alloy Value	129
	References	131
6	Energy-band Structure: Energy-band Gaps	133
6.1	Introductory Remarks	133
6.1.1	Quasi-cubic Band Model	133
6.1.2	Bowing Parameter	136
6.1.3	Ordered Alloy	136
6.2	Group-IV Semiconductor Alloy	139
6.2.1	Binary Alloy	139
	(a) CSi	139
	(b) CGe	140
	(c) SiGe	141
	(d) GeSn	144
6.2.2	Ternary Alloy	145
	(a) CSiGe	145
	(b) SiGeSn	147
6.2.3	Summary	147

6.3	III–V Semiconductor Ternary Alloy	149
6.3.1	(III, III)–N Alloy	149
	(a) <i>c</i> -AlGaN	149
	(b) <i>w</i> -AlGaN	150
	(c) <i>c</i> -AlInN	152
	(d) <i>w</i> -AlInN	152
	(e) <i>c</i> -GaInN	152
	(f) <i>w</i> -GaInN	152
6.3.2	(III, III)–P Alloy	153
	(a) AlGaP	153
	(b) AlInP	153
	(c) GaInP	155
6.3.3	(III, III)–As Alloy	157
	(a) AlGaAs	157
	(b) AlInAs	160
	(c) GaInAs	161
6.3.4	(III, III)–Sb Alloy	163
	(a) AlGaSb	163
	(b) AlInSb	165
	(c) GaInSb	166
6.3.5	Dilute-nitride III–(V, V) Alloy	166
	(a) General Considerations	166
	(b) GaNP	170
	(c) GaNAs	171
	(d) GaNSb	172
	(e) InNP	173
	(f) InNAs	173
	(g) InNSb	174
6.3.6	Al–(V, V) Alloy	174
6.3.7	Ga–(V, V) Alloy	175
	(a) GaPAs	175
	(b) GaPSb	176
	(c) GaAsSb	178
6.3.8	In–(V, V) Alloy	179
	(a) InPAs	179
	(b) InPSb	179
	(c) InAsSb	181
6.3.9	Summary	183
6.4	III–V Semiconductor Quaternary Alloy	184
6.4.1	Dilute-nitride Quaternary Alloy	184
	(a) AlGaNAs	184
	(b) GaInNP	185
	(c) GaInNAs	185
	(d) GaNPAs	185
6.4.2	(III, III)–(V, V) Alloy	186
	(a) AlGaPAs	186
	(b) AlGaAsSb	187

	(c) AlInAsSb	188
	(d) GaInPAs	189
	(e) GaInAsSb	191
6.4.3	(III, III, III)–V Alloy	193
	(a) AlGaInP	193
	(b) AlGaInAs	194
6.4.4	III–(V, V, V) Alloy	195
6.4.5	Summary	196
6.5	II–VI Semiconductor Alloy	197
6.5.1	(II, II)–O Ternary Alloy	197
	(a) BeZnO	197
	(b) MgZnO	197
	(c) ZnCdO	198
6.5.2	(II, II)–S Ternary Alloy	198
	(a) MgZnS	198
	(b) ZnCdS	200
	(c) ZnHgS	200
	(d) CdHgS	200
6.5.3	(II, II)–Se Ternary Alloy	201
	(a) BeZnSe	201
	(b) BeCdSe	202
	(c) MgZnSe	202
	(d) MgCdSe	202
	(e) ZnCdSe	203
	(f) CdHgSe	203
6.5.4	(II, II)–Te Ternary Alloy	204
	(a) BeZnTe	204
	(b) MgZnTe	205
	(c) MgCdTe	206
	(d) ZnCdTe	206
	(e) ZnHgTe	207
	(f) CdHgTe	208
6.5.5	Zn–(VI, VI) Ternary Alloy	208
	(a) ZnOS	208
	(b) ZnOSe	209
	(c) ZnSSe	209
	(d) ZnSTe	210
	(e) ZnSeTe	210
6.5.6	Cd–(VI, VI) Ternary Alloy	210
	(a) CdSSe	210
	(b) CdSTe	211
	(c) CdSeTe	211
6.5.7	(II, II)–(VI, VI) Quaternary Alloy	212
	(a) MgZnSSe	212
	(b) MgZnSeTe	213
6.5.8	(II, II, II)–VI Quaternary Alloy	213
	(a) BeMgZnSe	213

	(b) BeZnCdSe	213
	(c) MgZnCdSe	214
	6.5.9 Summary	214
	References	216
7	Energy-band Structure: Effective Masses	229
7.1	Introductory Remarks	229
7.1.1	Electron Effective Mass	229
7.1.2	Hole Effective Mass	230
7.1.3	Interpolation Scheme	232
7.2	Group-IV Semiconductor Alloy	234
7.2.1	CSi Binary Alloy	234
7.2.2	SiGe Binary Alloy	237
7.3	III–V Semiconductor Ternary Alloy	238
7.3.1	(III, III)–N Alloy	238
7.3.2	(III, III)–P Alloy	238
7.3.3	(III, III)–As Alloy	239
	(a) AlGaAs	239
	(b) AlInAs	241
	(c) GaInAs	241
7.3.4	(III, III)–Sb Alloy	242
7.3.5	Dilute-nitride III–(V, V) Alloy	243
	(a) Ga–(N, V) Alloy	243
	(b) In–(N, V) Alloy	244
7.3.6	Al–(V, V) Alloy	245
7.3.7	Ga–(V, V) Alloy	245
	(a) GaPAs	245
	(b) GaAsSb	246
7.3.8	In–(V, V) Alloy	246
	(a) InPAs	246
	(b) InAsSb	247
7.4	III–V Semiconductor Quaternary Alloy	248
7.4.1	Dilute-nitride Quaternary Alloy	248
7.4.2	(III, III)–(V, V) Alloy	248
	(a) GaInPAs	248
	(b) GaInAsSb	249
7.4.3	(III, III, III)–V Alloy	250
	(a) AlGaInP	250
	(b) AlGaInAs	250
7.4.4	III–(V, V, V) Alloy	250
7.5	II–VI Semiconductor Alloy	251
7.5.1	(II, II)–VI Ternary Alloy	251
7.5.2	II–(VI, VI) Ternary Alloy	253
7.5.3	(II, II)–(VI, VI) Quaternary Alloy	253
7.6	Concluding Remarks	254
7.6.1	Composition Dependence	254
7.6.2	External Perturbation Effect	254
	References	256

8 Deformation Potentials	259
8.1 Intravalley Deformation Potential: Γ Point	259
8.1.1 Group-IV Semiconductor Alloy	259
8.1.2 III–V Semiconductor Ternary Alloy	261
(a) AlGaN	261
(b) GaInN	262
(c) AlInP	262
(d) GaInP	263
(e) AlGaAs	263
(f) AlInAs	264
(g) GaInAs	265
(h) GaNAs	265
(i) GaPAs	266
8.1.3 III–V Semiconductor Quaternary Alloy	266
8.1.4 II–VI Semiconductor Alloy	266
8.2 Intravalley Deformation Potential: High-symmetry Points	267
8.2.1 Group-IV Semiconductor Alloy	267
8.2.2 III–V Semiconductor Alloy	269
8.2.3 II–VI Semiconductor Alloy	270
8.3 Intervalley Deformation Potential	270
8.3.1 Group-IV Semiconductor Alloy	270
8.3.2 III–V Semiconductor Alloy	271
8.3.3 II–VI Semiconductor Alloy	272
References	272
9 Heterojunction Band Offsets and Schottky Barrier Height	275
9.1 Heterojunction Band Offsets	275
9.1.1 General Considerations	275
9.1.2 Group-IV Semiconductor Heterostructure System	275
(a) CSi/Si	275
(b) SiGe/Si	276
(c) CSiGe/Si	276
(d) CSi/SiGe	277
9.1.3 III–V Semiconductor Heterostructure System: Lattice-matched Ternary-alloy System	277
(a) GaInP/GaAs	277
(b) GaInP/AlInP	277
(c) GaInP/AlGaAs	278
(d) AlGaAs/GaAs	278
(e) AlInAs/InP	279
(f) GaInAs/InP	279
(g) GaInAs/AlInAs	280
(h) InAsSb/GaSb	280
9.1.4 III–V Semiconductor Heterostructure System: Lattice-matched Quaternary Alloy	280
(a) GaInPAs/InP	280

	(b) AlGaAsSb/GaSb	281
	(c) AlGaAsSb/InP	281
	(d) AlGaAsSb/InAs	281
	(e) GaInAsSb/GaSb	281
	(f) GaInAsSb/InP	281
	(g) GaInAsSb/InAs	281
	(h) AlGaAsSb/GaInAsSb	282
	(i) AlInAsSb/GaInAsSb	282
	(j) AlGaInP/AlInP	282
	(k) AlGaInP/GaInP	282
	(l) AlGaInAs/GaInAs	283
9.1.5	III–V Semiconductor Heterostructure System: Lattice-mismatched Alloy System	283
	(a) w -AlGaN/ w -GaN	283
	(b) w -AlInN/ w -InN	284
	(c) w -GaInN/ w -GaN	284
	(d) GaInAs/GaAs	284
	(e) Dilute-nitride-based Heterostructure System	286
9.1.6	II–VI Semiconductor Heterostructure System	287
	(a) (II, II)–O-based Heterostructure System	287
	(b) (II, II)–S-based Heterostructure System	287
	(c) (II, II)–Se-based Heterostructure System	287
	(d) (II, II)–Te-based Heterostructure System	288
	(e) Zn–(VI, VI)-based Heterostructure System	289
	(f) (II, II)–VI-based and II–(VI, VI)-based Heterostructure Systems	289
	(g) (II, II)–(VI, VI)-based Heterostructure System	289
	(h) (II, II, II)–VI-based Heterostructure System	289
9.2	Schottky Barrier Height	289
9.2.1	General Considerations	289
9.2.2	Group-IV Semiconductor Alloy	290
	(a) SiGe Binary Alloy	290
	(b) CSiGe Ternary Alloy	291
9.2.3	III–V Semiconductor Ternary Alloy	291
	(a) (III, III)–N Alloy	291
	(b) (III, III)–P Alloy	292
	(c) (III, III)–As Alloy	292
	(d) (III, III)–Sb Alloy	294
	(e) Ga–(V, V) Alloy	295
9.2.4	III–V Semiconductor Quaternary Alloy	296
	(a) (III, III)–(V, V) Alloy	296
	(b) (III, III, III)–V Alloy	297
9.2.5	II–VI Semiconductor Alloy	299
	(a) (II, II)–Te Ternary Alloy	299
	(b) Zn–(VI, VI) Ternary Alloy	300
	References	300

10 Optical Properties	307
10.1 Introductory Remarks	307
10.1.1 Optical Dispersion Relations	307
10.1.2 Static and High-frequency Dielectric Constants	308
10.2 Group-IV Semiconductor Alloy	308
10.2.1 Binary Alloy	308
(a) CSi	308
(b) CGe	309
(c) SiGe	309
(d) GeSn	310
10.2.2 Ternary Alloy	311
10.3 III–V Semiconductor Ternary Alloy	311
10.3.1 (III, III)–N Alloy	311
(a) <i>c</i> -(III, III)–N Alloy	311
(b) <i>w</i> -AlGa _N	312
(c) <i>w</i> -AlIn _N	313
(d) <i>w</i> -GaIn _N	313
10.3.2 (III, III)–P Alloy	314
(a) AlGaP	314
(b) AlInP	314
(c) GaInP	315
10.3.3 (III, III)–As Alloy	316
(a) AlGaAs	316
(b) AlInAs	316
(c) GaInAs	318
10.3.4 (III, III)–Sb Alloy	319
(a) AlGaSb	319
(b) GaInSb	319
10.3.5 Dilute-nitride III–(V, V) Alloy	320
(a) GaNP	320
(b) GaNAs	321
(c) GaNSb	321
(d) InNP	322
(e) InNAs	322
10.3.6 Al–(V, V) Alloy	322
10.3.7 Ga–(V, V) Alloy	323
(a) GaPAs	323
(b) GaPSb	324
(c) GaAsSb	324
10.3.8 In–(V, V) Alloy	324
(a) InPAs	324
(b) InPSb	325
(c) InAsSb	325
10.4 III–V Semiconductor Quaternary Alloy	326
10.4.1 Dilute-nitride Quaternary Alloy	326
(a) GaInNP	326
(b) GaInNAs	326

10.4.2	(III, III)–(V, V) Alloy	326
	(a) AlGaPAs	326
	(b) AlGaAsSb	327
	(c) GaInPAs	327
	(d) GaInAsSb	328
10.4.3	(III, III, III)–V Alloy	330
	(a) AlGaInP	330
	(b) AlGaInAs	331
10.4.4	III–(V, V, V) Alloy	332
10.5	II–VI Semiconductor Alloy	332
10.5.1	(II, II)–O Ternary Alloy	332
	(a) BeZnO	332
	(b) MgZnO	332
	(c) ZnCdO	333
10.5.2	(II, II)–S Ternary Alloy	333
	(a) MgZnS	333
	(b) ZnCdS	334
10.5.3	(II, II)–Se Ternary Alloy	335
	(a) BeZnSe	335
	(b) BeCdSe	335
	(c) MgZnSe	335
	(d) MgCdSe	337
	(e) ZnCdSe	337
	(f) CdHgSe	338
10.5.4	(II, II)–Te Ternary Alloy	339
	(a) BeZnTe	339
	(b) MgZnTe	339
	(c) MgCdTe	340
	(d) ZnCdTe	340
	(e) ZnHgTe	342
	(f) CdHgTe	342
10.5.5	Zn–(VI, VI) Ternary Alloy	343
	(a) ZnOS	343
	(b) ZnSSe	343
	(c) ZnSTe	343
	(d) ZnSeTe	344
10.5.6	Cd–(VI, VI) Ternary Alloy	344
	(a) CdSSe	344
	(b) CdSTe	345
	(c) CdSeTe	346
10.5.7	(II, II)–(VI, VI) Quaternary Alloy	346
	(a) MgZnSSe	346
	(b) MgZnSeTe	346
10.5.8	(II, II, II)–VI Quaternary Alloy	346
	(a) BeMgZnSe	346
	(b) BeZnCdSe	346
	(c) MgZnCdSe	346
	References	347

11 Elasto-optic, Electro-optic and Nonlinear Optical Properties	357
11.1 Elasto-optic Effect	357
11.1.1 Group-IV Semiconductor Alloy	357
11.1.2 III–V Semiconductor Alloy	357
11.1.3 II–VI Semiconductor Alloy	357
11.2 Linear Electro-optic Constant	358
11.2.1 Group-IV Semiconductor Alloy	358
11.2.2 III–V Semiconductor Alloy	358
11.2.3 II–VI Semiconductor Alloy	358
11.3 Quadratic Electro-optic Constant	360
11.3.1 Group-IV Semiconductor Alloy	360
11.3.2 III–V Semiconductor Alloy	360
11.3.3 II–VI Semiconductor Alloy	360
11.4 Franz–Keldysh Effect	360
11.4.1 Group-IV Semiconductor Alloy	360
11.4.2 III–V Semiconductor Alloy	360
11.4.3 II–VI Semiconductor Alloy	360
11.5 Nonlinear Optical Constant	361
11.5.1 Group-IV Semiconductor Alloy	361
11.5.2 III–V Semiconductor Alloy	361
11.5.3 II–VI Semiconductor Alloy	362
References	362
 12 Carrier Transport Properties	 365
12.1 Introductory Remarks	365
12.2 Low-field Mobility	367
12.2.1 Group-IV Semiconductor Alloy	367
(a) CSi Binary Alloy	367
(b) SiGe Binary Alloy	367
12.2.2 III–V Semiconductor Ternary Alloy	368
(a) (III, III)–N Alloy	368
(b) (III, III)–P Alloy	369
(c) (III, III)–As Alloy	369
(d) (III, III)–Sb Alloy	369
(e) Dilute-nitride III–(V, V) Alloy	371
(f) Ga–(V, V) Alloy	371
(g) In–(V, V) Alloy	371
12.2.3 III–V Semiconductor Quaternary Alloy	371
(a) Dilute-nitride Alloy	371
(b) (III, III)–(V, V) Alloy	372
(c) (III, III, III)–V Alloy	373
12.2.4 II–VI Semiconductor Alloy	374
(a) (II, II)–O Ternary Alloy	374
(b) (II, II)–Se Ternary Alloy	374
(c) (II, II)–Te Ternary Alloy	375
(d) Zn–(V, V) Ternary Alloy	376

12.3	High-field Transport	376
12.3.1	Group-IV Semiconductor Alloy	376
12.3.2	III–V Semiconductor Ternary Alloy	376
	(a) (III, III)–N Alloy	376
	(b) (III, III)–P Alloy	376
	(c) (III, III)–As Alloy	377
	(d) Dilute-nitride III–(V, V) Alloy	378
12.3.3	III–V Semiconductor Quaternary Alloy	378
12.3.4	II–VI Semiconductor Alloy	379
12.4	Minority-carrier Transport	379
12.4.1	Group-IV Semiconductor Alloy	379
	(a) SiGe Binary Alloy	379
	(b) CSiGe Ternary Alloy	380
12.4.2	III–V Semiconductor Ternary Alloy	380
	(a) (III, III)–N Alloy	380
	(b) (III, III)–As Alloy	380
	(c) Ga–(V, V) Alloy	381
12.4.3	III–V Semiconductor Quaternary Alloy	381
	(a) (III, III)–(V, V) Alloy	381
	(b) (III, III, III)–V Alloy	382
12.4.4	II–VI Semiconductor Alloy	382
12.5	Impact Ionization Coefficient	382
12.5.1	Group-IV Semiconductor Alloy	382
12.5.2	III–V Semiconductor Ternary Alloy	382
	(a) (III, III)–N Alloy	382
	(b) (III, III)–P Alloy	383
	(c) (III, III)–As Alloy	383
	(d) (III, III)–Sb Alloy	384
	(e) Dilute-nitride III–(V, V) Alloy	384
	(f) Ga–(V, V) Alloy	385
	(g) In–(V, V) Alloy	385
12.5.3	III–V Semiconductor Quaternary Alloy	385
	(a) (III, III)–(V, V) Alloy	385
	(b) (III, III, III)–V Alloy	385
12.5.4	II–VI Semiconductor Alloy	386
	References	386

Series Preface

WILEY SERIES IN MATERIALS FOR ELECTRONIC AND OPTOELECTRONIC APPLICATIONS

This book series is devoted to the rapidly developing class of materials used for electronic and optoelectronic applications. It is designed to provide much-needed information on the fundamental scientific principles of these materials, together with how these are employed in technological applications. The books are aimed at (postgraduate) students, researchers and technologists, engaged in research, development and the study of materials in electronics and photonics, and industrial scientists developing new materials, devices and circuits for the electronic, optoelectronic and communications industries.

The development of new electronic and optoelectronic materials depends not only on materials engineering at a practical level, but also on a clear understanding of the properties of materials, and the fundamental science behind these properties. It is the properties of a material that eventually determine its usefulness in an application. The series therefore also includes such titles as electrical conduction in solids, optical properties, thermal properties, and so on, all with applications and examples of materials in electronics and optoelectronics. The characterization of materials is also covered within the series in as much as it is impossible to develop new materials without the proper characterization of their structure and properties. Structure-property relationships have always been fundamentally and intrinsically important to materials science and engineering.

Materials science is well known for being one of the most interdisciplinary sciences. It is the interdisciplinary aspect of materials science that has led to many exciting discoveries, new materials and new applications. It is not unusual to find scientists with a chemical engineering background working on materials projects with applications in electronics. In selecting titles for the series, we have tried to maintain the interdisciplinary aspect of the field, and hence its excitement to researchers in this field.

Peter Capper
Safa Kasap
Arthur Willoughby

Preface

Semiconductor alloys provide a natural means of tuning the magnitude of the band-gap energy and other material properties so as to optimize and widen the application of semiconductor devices. Current research and development in semiconductor alloys is focused on areas of improved materials growth, development of unique materials characterization and suitable process technologies, fabrication of novel devices using artificially controlled materials structures and better understanding of degradation mechanisms in electronic and optoelectronic devices for improved device reliability.

Even though the basic semiconductor alloy concepts are understood at this time, the determination of some device parameters has been hampered by a lack of definite knowledge of many material parameters. The main purpose of this book is to provide a comprehensive treatment of the materials aspects of group-IV, III–V and II–VI semiconductor alloys used in various electronic and optoelectronic devices. The topics treated in this book include the structural, thermal, mechanical, lattice vibronic, electronic, optical and carrier transport properties of such semiconductor alloys. The book covers not only commonly known alloys (SiGe, AlGaAs, GaInPAs, ZnCdTe, etc.) but also new alloys, such as dilute-carbon alloys (CSiGe, CSiSn, etc.), III–N alloys, dilute-nitride alloys (GaNAs, GaInNAs, etc.) and Mg- or Be-based II–VI semiconductor alloys.

The reader may also find the companion book '*Properties of Group-IV, III–V and II–VI Semiconductors*' published in this series useful since it emphasizes the endpoint semiconductor principles and properties.

The extensive bibliography is included for those who wish to find additional information if required. It is hoped that the book will attract the attention of not only semiconductor device engineers, but also solid-state physicists and materials scientists, and particularly postgraduate students, R&D staff and teaching and research professionals.

Sadao Adachi,
Gunma University, Japan

Abbreviations and Acronyms

BAC	band anticrossing
bct	body-centered tetragonal
BZ	Brillouin zone
<i>c</i> -	cubic-
CB	conduction band
CL	cathodoluminescence
Cl-VPE	chloride-transport vapor-phase epitaxy
CP	critical point
CR	cyclotron resonance
<i>C</i> - <i>V</i>	capacitance-voltage
CVD	chemical vapor deposition
DLTS	deep-level transient spectroscopy
DOS	density of states
DP	deformation potential
ER	electroreflectance
EXAFS	extended X-ray absorption fine-structure spectroscopy
fcc	face-centered cubic
FET	field effect transistor
FTIR	Fourier transform infrared
HBT	heterojunction bipolar transistor
HH	heavy-hole
HREELS	high-resolution electron-energy-loss spectroscopy
HRXRD	high-resolution X-ray diffraction
HT-VPE	hydride-transport vapor-phase epitaxy
IR	infrared
<i>I</i> - <i>V</i>	current-voltage
LA	longitudinal acoustic
LH	light-hole
LO	longitudinal optical
LPE	liquid-phase epitaxy
MBE	molecular beam epitaxy
MDF	model dielectric function
MOCVD	metalorganic chemical vapor deposition

MOMBE	metalorganic molecular beam epitaxy
MQW	multiple quantum well
MREI	modified random-element-isodisplacement
PDP	phonon deformation potential
PIXE	particle-induced X-ray emission
PL	photoluminescence
PLD	pulsed-laser deposition
PLE	photoluminescence excitation
PR	photorefectance
PzR	piezorefectance
QW	quantum well
RBS	Rutherford backscattering spectrometry
rf	radio-frequency
RHEED	reflection high-energy electron diffraction
SE	spectroscopic ellipsometry
SL	superlattice
SO	spin-orbit
TA	transverse acoustic
TEM	transmission electron microscopy
THM	traveling heater method
TO	transverse optical
TP	triple-period
TR	thermoreflectance
UHV	ultrahigh vacuum
UV	ultraviolet
VB	valence band
VCA	virtual crystal approximation
VPE	vapor-phase epitaxy
w-	wurtzite
WDXS	wavelength-dispersive X-ray spectroscopy
XPS	X-ray photoelectron spectroscopy
XRD	X-ray diffraction
1D	one-dimensional
2D	two-dimensional
3D	three-dimensional

Introductory Remarks

A.1 AN ALLOY AND A COMPOUND

An alloy is a combination, either in solution or compound, of two or more elements. An alloy with two components is called a binary alloy; one with three is a ternary alloy; one with four is a quaternary alloy; one with five is a pentanary alloy. The resulting alloy substance generally has properties significantly different from those of its components. The proportions of the ingredients are available.

A chemical compound is a substance consisting of two or more chemical elements that are chemically combined in fixed proportions. The ratio of each element is usually expressed by chemical formula. For example, water is a compound consisting of two hydrogen atoms bonded to an oxygen atom (H_2O). The atoms within a compound can be held together by a variety of interactions, ranging from covalent bonds to electrostatic forces in ionic bonds. A continuum of bond polarities exists between the purely covalent and ionic bonds. For example, H_2O is held together by polar covalent bonds. NaCl is an example of an ionic compound.

Simply, an alloy is formed from a physical mixture of two or more substances, while a compound is formed from a chemical reaction. An alloy crystal is sometimes called a mixed crystal or a solid solution. For example, GaAs is a compound consisting of Ga atoms bonded to As atoms. It is not an alloy. $\text{Al}_x\text{Ga}_{1-x}\text{As}$ is an alloy compound consisting of AlAs and GaAs with a mole ratio of $x:(1-x)$. The bonds in GaAs and AlAs are not adequately described by any of these extreme types, but have characteristics intermediate to those usually associated with the covalent and ionic terms. The bonds in diamond, $\text{C}-\text{C}$, can be described by the covalent bond term only. It is an elemental semiconductor, not a compound semiconductor. Similarly, Si and Ge are elemental semiconductors. Like $\text{Al}_x\text{Ga}_{1-x}\text{As}$, $\text{Si}_x\text{Ge}_{1-x}$ ($0 \leq x \leq 1.0$) is an alloy semiconductor. The bonds $\text{Si}-\text{Ge}$, $\text{Si}-\text{Si}$ and $\text{Ge}-\text{Ge}$ in $\text{Si}_x\text{Ge}_{1-x}$ are, therefore, described by the covalent term only. It should be noted, however, that silicon carbide (SiC) is a compound, not an alloy. This is because that the chemical bonds in SiC cannot be described only by the covalent term, but have characteristics intermediate to those associated with the covalent and ionic terms, like GaAs and AlAs .

There is an ordered alloy phase in $\text{Si}_x\text{Ge}_{1-x}$ binary alloy [1]. This phase exhibits long-range order rather than random arrangement of atoms as expected previously. The ordered phase, as expected, occurs mostly in bulk $\text{Si}_x\text{Ge}_{1-x}$ layers at $x \sim 0.5$ and can be explained by the rhombohedral structure. Note that SiC is thought to be an ordered alloy. It crystallizes in a large number of polytypes. The various types of SiC differ one from another only by the order in which successive planes of Si (or C) atoms are stacked along the c axis; one polytype is the cubic, zinc-blende structure (3C) while the reminder, including two of the more frequently occurring forms, 6H (hexagonal) and 15R (rhombohedral), possess uniaxial symmetry [2]. There is no diamond structure in the SiC polytypes. An ordered alloy phase has been found not only in $\text{Si}_x\text{Ge}_{1-x}$, but also in many III–V and II–VI semiconductor alloys [3].

A.2 GRIMM-SOMMERFELD RULE

The periodic law is most commonly expressed in chemistry in the form of a periodic table or chart. An important part of this table is shown in Table A.1. The so-called short-form periodic table, based on the Mendeleyev table, with subsequent emendations and additions, is still in widespread use. The elements in this table are arranged in seven horizontal rows, called the periods, in order of increasing atomic weights, and in 18 vertical columns, called the groups. The first period, containing two elements, H and He, and the next two periods, each containing eight elements, are called the short periods. The remaining periods, called the long periods, contain 18 elements, as in periods 4 and 5, or 32 elements, as in period 6. The long period 7 includes the actinide series, which has been filled in by the synthesis of radioactive nuclei through element 103, lawrencium (Lr). Heavier transuranium elements, atomic numbers 104 to 112, have also been synthesized.

Table A.1 Periodic table

	Ia (1)									VIIIa (18)
1	1 H	IIa (2)		IIIa (13)	IVa (14)	Va (15)	VIa (16)	VIIa (17)	2 He	
2	3 Li	4 Be		5 B	6 C	7 N	8 O	9 F	10 Ne	
3	11 Na	12 Mg		IIb (12)	13 Al	14 Si	15 P	16 S	17 Cl	18 Ar
4	19 K	20 Ca		30 Zn	31 Ga	32 Ge	33 As	34 Se	35 Br	36 Kr
5	37 Rb	38 Sr		48 Cd	49 In	50 Sn	51 Sb	52 Te	53 I	54 Xe
6	55 Cs	56 Ba		80 Hg	81 Tl	82 Pb	83 Bi	84 Po	85 At	86 Rn
7	87 Fr									118 Uuo

The elemental semiconductors in column IVa of the periodic table are diamond, Si and Ge, and they are of some importance in various device applications. Note that gray tin (α -Sn) is a semiconductor (semimetal) and Pb is a metal. SiC is the only compound semiconductor formed by column IVa elements. The III–V and II–VI semiconductors possess a crystal structure similar to either one of the cubic mineral, sphalerite or to hexagonal wurtzite. There are 15 III–V, 18 II–VI and four I–VII compound semiconductor. The four I–VII compound semiconductors are: CuCl, CuBr, CuI and AgI.

A substantial development in the search for semiconducting materials with new combinations of physical and chemical properties was reached when Goryunova using the ideas of Huggins [4] and Grimm and Sommerfeld [5], developed a method of prediction of the composition of chemical compounds with the tetrahedral and octahedral (NaCl-type) coordination of atoms in their crystal lattice [6]. His method is based on consideration of the number of valence electrons in the elements, which is assumed to be equal to the number of the group in

the periodic table to which they belong (so-called full or normal valency) and can be expressed in the form of the equation system [7]

$$\begin{aligned} \sum_{i=1}^A B_i x_i &= 4 \\ \sum_{i=1}^n B_i x_i &= \sum_{i=n+1}^A (8-B_i) x_i \\ \sum_{i=1}^A x_i &= 1 \end{aligned} \quad (\text{A.1})$$

for calculating the content of the representatives of different groups of the periodic table in a compound containing A components, n of which are considered as cations, where B_i is the number of the group in the periodic table to which the i th component of the compounds belongs, and x_i is its concentration in the compound. The first expression in Equation (A.1) represents the condition that the average number of valence electrons per atom is equal to four (tetrahedral rule). The second equation states that in the compound, the number of valence electrons, which ‘cations’ give to ‘anions’ is equal to the number of electrons the anion needs to form the octet (i.e. the normal valency condition). The quotation marks here are used to remind us that in the solids with dominating covalent bonds, which factually form the class of semiconductor, the ion concept is quite conditional. Based on these criteria, it is possible to obtain all probable types of tetrahedral substances, taking into consideration only the location of their components in one or another group of the periodic table. Table A.2 briefly summarizes these results.

Table A.2 Summary of valence binary compounds and ternary analogs of binary valence compounds which satisfy tetrahedral rule ($Z=4$). ch=chalcopyrite; zb=zinc-blende; t=tetragonal; or=orthorhombic; w=wurtzite; h=hexagonal ($P6_3mc$ (C_{6v})); rh=rhombohedral

Cation	Compound	Typical compound (Crystal structure)
I	I–III–VI ₂	CuGaSe ₂ (ch)
	I–IV ₂ –V ₃	CuGe ₂ P ₃ (zb)
	I ₂ –IV–VI ₃	Cu ₂ GeSe ₃ (t)
	I ₃ –V–VI ₄	Cu ₃ PS ₄ (or)
	I–VII	γ-CuCl (w)
II	II ₄ –III–VII ₃	ZnSiP ₂ (ch)
	II–IV–V ₂	
	II ₃ –IV–VII ₂	
	II ₂ –V–VII	
	II–VI	CdTe (zb)
III	III ₂ –IV–VI	Al ₂ CO (w)
	III ₃ –IV ₂ –VII	GaAs (zb)
	III–V	
IV	IV–IV	SiC (zb, w, h, rh)

Table A.3 All the possible types of valence binary compound

Cation	Cation to anion content ratio x/y , (Z), Typical compound					
	II	III	IV	V	VI	VII
I	6:1 (1.14)	5:1 (1.33)	4:1 (1.60)	3:1 (2.00) K ₃ Sb	2:1 (2.67) Ag ₂ S	1:1 (4.00) γ -CuCl
II	—	5:2 (2.29)	2:1 (2.67) Mg ₂ Si	3:2 (3.20) Cd ₃ As ₂	1:1 (4.00) CdTe	1:2 (5.33) HgI ₂
III	—	—	4:3 (3.43)	1:1 (4.00) GaAs	2:3 (4.80) Ga ₂ Se ₃	1:3 (6.00)
IV	—	—	1:1 (4.00) SiC	3:4 (4.57)	1:2 (5.33) SnS ₂	1:4 (6.40)
V	—	—	—	—	2:5 (5.71)	1:5 (6.67)
VI	—	—	—	—	—	1:6 (6.86)

In analyzing the normal valency compounds, not only tetrahedral phases but all compounds of this category can be grouped, first of all, in accordance with the average valence electron content per atom of a compound, Z . We list in Table A.3 all possible types of the binary valence compounds [7]. If A and B are the cation valencies in a binary compound, the atomic ratio of components in a valence compound A_xB_y is given by

$$\frac{x}{y} = \frac{8-B}{A} \quad (\text{A.2})$$

and the valence electron concentration per atom in the corresponding binary compound is

$$Z = \frac{8A}{8-(B-A)} = \frac{8A}{8+(A-B)} \quad (\text{A.3})$$

as following from Equation (A.1).

In Table A.3, there are 22 types of binary valence compounds. Goryunova [6] concluded from the available experimental data that the phases possessing semiconducting properties are those with Z between two and six; the phases smaller than two are either metal alloys or the compounds with typical metallic crystal structure and properties. Goryunova also showed that there are 148 types of ternary compounds which are the electron analogs of the binary valence compounds [7]. Among them, there are 10 types of ternary compounds which satisfy the tetrahedral rule $Z=4$, as listed in Table A.2.

A.3 AN INTERPOLATION SCHEME

The electronic energy-band parameters of semiconductor alloys and their dependence on alloy composition are very important, and so they have received much attention in the past. Investigation

of many device parameters have, however, been hampered by a lack of definite knowledge of various material parameters. This necessitates the use of some sort of interpolation scheme.

If the linear interpolation scheme is used, the ternary material parameter T for an alloy of the form $A_xB_{1-x}C$ can be derived from binary parameters (B) by

$$T(x) = xB_{AC} + (1-x)B_{BC} \equiv a + bx \quad (\text{A.4})$$

where $a \equiv B_{BC}$ and $b \equiv B_{AC} - B_{BC}$. Some material parameters, however, deviate significantly from the linear relationship shown in Equation (A.4), and exhibit an approximately quadratic dependence on x . The ternary material parameter in such a case can be very efficiently approximated by the relationship

$$T(x) = xB_{AC} + (1-x)B_{BC} + x(1-x)C_{AB} \equiv a + bx + cx^2 \quad (\text{A.5})$$

where $a \equiv B_{BC}$, $b \equiv B_{AC} - B_{BC} + C_{AB}$ and $c \equiv -C_{AB}$. The parameter c is called a bowing parameter.

The quaternary material $A_xB_{1-x}C_yD_{1-y}$ is thought to be constructed from four binaries: AC, AD, BC and BD. If the linear interpolation scheme is used, the quaternary material parameter Q can be derived from the binary parameters by

$$Q(x, y) = xyB_{AC} + x(1-y)B_{AD} + (1-x)yB_{BC} + (1-x)(1-y)B_{BD} \quad (\text{A.6})$$

If one of the four binary parameters (e.g. B_{AD}) is lacking, Q can be estimated from

$$Q(x, y) = xB_{AC} + (y-x)B_{BC} + (1-y)B_{BD} \quad (\text{A.7})$$

The quaternary material $A_xB_yC_{1-x-y}D$ is thought to be constructed from three binaries: AD, BD and CD. The corresponding linear interpolation is given by

$$Q(x, y) = xB_{AD} + yB_{BD} + (1-x-y)B_{CD} \quad (\text{A.8})$$

If relationships for the ternary parameters are available, the quaternary material parameter Q can be expressed either as $(A_xB_{1-x}C_yD_{1-y})$

$$Q(x, y) = \frac{x(1-x)[yT_{ABC}(x) + (1-y)T_{ABD}(x)] + y(1-y)[xT_{ACD}(y) + (1-x)T_{BCD}(y)]}{x(1-x) + y(1-y)} \quad (\text{A.9})$$

or $(A_xB_yC_{1-x-y}D)$

$$Q(x, y) = \frac{xyT_{ABD}(u) + y(1-x-y)T_{BCD}(v) + x(1-x-y)T_{ACD}(w)}{xy + y(1-x-y) + x(1-x-y)} \quad (\text{A.10})$$

with

$$\begin{aligned} u &= (1-x-y)/2 \\ v &= (2-x-2y)/2 \\ w &= (2-2x-y)/2 \end{aligned} \quad (\text{A.11})$$

If a quaternary bowing term $D_{ABCD} = -d$ is considered in Equation (A.9) or Equation (A.10), we can obtain

$$Q(x, y) = \{x(1-x)[yT_{ABC}(x) + (1-y)T_{ABD}(x) + y(1-y)D_{ABCD}] + y(1-y) \times [xT_{ACD}(y) + (1-x)T_{BCD}(y) + x(1-x)D_{ABCD}]\} \times [x(1-x) + y(1-y)]^{-1} \quad (\text{A.12})$$

for the $A_xB_{1-x}C_yD_{1-y}$ quaternary or

$$Q(x, y) = \frac{xyT_{ABD}(u) + y(1-x-y)T_{BCD}(v) + x(1-x-y)T_{ACD}(w) + xy(1-x-y)D_{ABCD}}{xy + y(1-x-y) + x(1-x-y)} \quad (\text{A.13})$$

for the $A_xB_yC_{1-x-y}D$ quaternary.

The weighted form of Equation (A.12) can be written as

$$\begin{aligned} Q(x, y) = & y(1-x)B_{BC} + xyB_{AC} + (1-y)xB_{AD} + (1-x)(1-y)B_{CD} + x(1-x)(1-y)C_{AB(D)} \\ & + x(1-x)yC_{AB(C)} + (1-x)y(1-y)C_{(B)CD} + xy(1-y)C_{(A)CD} \\ & + x(1-x)y(1-y)D_{ABCD} \end{aligned} \quad (\text{A.14})$$

where $C_{AB(D)}$ is the ternary bowing parameter for an alloy $A_xB_{1-x}D$ and so on. Expression (A.14) can be conveniently compacted into an inner product whose 3×3 matrix includes only the alloy parameters and closely resembles the map of the quaternary alloy in its compositional space [8]

$$Q(x, y) = [y \ y(1-y) \ 1-y] \begin{bmatrix} B_{BC} & C_{AB(C)} & B_{AC} \\ C_{(B)CD} & D_{ABCD} & C_{(A)CD} \\ B_{BD} & C_{AB(D)} & B_{AD} \end{bmatrix} \begin{bmatrix} 1-x \\ x(1-x) \\ x \end{bmatrix} \quad (\text{A.15})$$

If the bowing parameters C and D in Equation (A.15) are zero, Vegard law is recovered in its bilinear formulation. Equation (A.15) is valid for quaternaries of types $A_xB_{1-x}C_yD_{1-y}$ and $A_xB_yC_{1-x-y}D$ ($AB_xC_yD_{1-x-y}$). For the latter, one binary can be assigned twice in the same row or column of the alloy matrix with a zero bowing factor (i.e. $B_{AC} = B_{AD}$ and $C_{(A)CD} = 0$). The composition for such an alloy is given by $(B_{AD})_x[(B_{BD})_y(B_{CD})_{1-y}]_{1-x}$ (e.g., $Al_x(Ga_yIn_{1-y})_{1-x}P$).

The essentially same expressions can be obtained for group-IV semiconductor alloys. The binary material parameter B in the form of A_xB_{1-x} can be written using the elemental material parameters (A) as

$$B(x) = xA_A + (1-x)A_B \equiv a + bx \quad (\text{A.16})$$

The bowing effect modifies Equation (A.16) in the form

$$B(x) = xA_A + (1-x)A_B + x(1-x)C_{AB} \equiv a + bx + cx^2 \quad (\text{A.17})$$

Similarly, the ternary material parameter T in $A_xB_yC_{1-x-y}$ can be expressed as

$$T(x, y) = xA_A + yA_B + (1-x-y)A_C \quad (\text{A.18})$$